

SQL

PALAKURTHI LALITH PRAKASH

2303A52012

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```
CREATE DATABASE company_db;
```

```
USE company_db;
```

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    department VARCHAR(50),  
    salary INT,  
    hire_date DATE  
);
```

```
INSERT INTO employees  
(emp_id, emp_name, department, salary, hire_date)  
VALUES  
(1, 'Arun', 'IT', 90000, '2024-01-10'),  
(2, 'Bala', 'IT', 80000, '2024-02-15'),  
(3, 'Charan', 'IT', 80000, '2024-03-20'),  
(4, 'Divya', 'HR', 75000, '2024-01-12'),  
(5, 'Esha', 'HR', 70000, '2024-04-01'),  
(6, 'Farhan', 'HR', 70000, '2024-05-05'),  
(7, 'Gokul', 'Sales', 95000, '2024-02-02'),  
(8, 'Hari', 'Sales', 85000, '2024-06-18');
```

```
SELECT * FROM employees;
```

```
-- 1.
```

```
SELECT ROW_NUMBER() OVER ( ORDER BY salary DESC ) AS row_num ,emp_id,  
emp_name, department, salary FROM employees;
```

```
-- 2.
```

```
SELECT RANK() OVER ( ORDER BY salary DESC ) AS row_num ,emp_id, emp_name,  
department, salary FROM employees;
```

```
-- 3.
```

```
SELECT DENSE_RANK() OVER ( ORDER BY salary DESC ) AS row_num ,emp_id,  
emp_name, department, salary FROM employees;
```

```
-- 4.
```

```
SELECT ROW_NUMBER() OVER ( PARTITION BY department ORDER BY salary DESC ) AS  
row_num, emp_name, department, salary FROM employees;
```

-- 5.

```
SELECT RANK() OVER ( PARTITION BY department ORDER BY salary DESC ) AS  
salary_rank, emp_name, department, salary FROM employees;
```

-- 6.

```
SELECT * FROM (  
    SELECT emp_id, emp_name, department, salary, DENSE_RANK() OVER ( PARTITION BY  
department ORDER BY salary DESC ) AS salary_rank FROM employees  
) AS x
```

```
WHERE salary_rank <= 2;
```

-- 7.

```
SELECT emp_name, department, salary, FIRST_VALUE(emp_name) OVER ( PARTITION BY  
department ORDER BY salary DESC ) AS highest_paid_employee FROM employees;
```

-- 8.

```
SELECT emp_name, department, salary, LAST_VALUE(emp_name) OVER ( PARTITION BY  
department ORDER BY salary DESC ROWS BETWEEN UNBOUNDED PRECEDING AND  
UNBOUNDED FOLLOWING ) AS lowest_paid_employee
```

```
FROM employees;
```

-- 9.

```
SELECT emp_name, department, salary,  
    FIRST_VALUE(salary) OVER ( PARTITION BY department ORDER BY salary DESC ) AS  
highest_salary,  
    LAST_VALUE(salary) OVER ( PARTITION BY department ORDER BY salary DESC ROWS  
BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING ) AS lowest_salary  
FROM employees;
```

-- 10.

```
SELECT * FROM (  
    SELECT emp_id, emp_name, department, salary,  
        DENSE_RANK() OVER ( ORDER BY salary DESC ) AS salary_rank FROM employees )
```

```
AS x
```

```
WHERE salary_rank = 2;
```